

NYC Taxi Demand Forecasting

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Abstract—This paper examines deep learning methods for predicting hourly taxi demand across New York City zones. The study utilizes 35.3 million taxi trip records from 2022-2024 to build and compare different forecasting models, including LSTM, GRU, Temporal CNN, and Transformer networks. The approach includes feature engineering, hyperparameter optimization, and ensemble methods. RandomForest achieved the best performance with a Mean Absolute Error (MAE) of 5.56 trips/hour and MAPE of 10.0%, while the stacked ensemble was the best deep learning approach at 6.96 MAE. Results demonstrate that well-engineered features with tree-based methods can outperform complex neural networks for urban transportation demand forecasting.

I. INTRODUCTION

Accurate prediction of taxi demand is vital for improving urban transportation systems, reducing passenger wait times, and increasing efficiency for ride-sharing companies. The main challenge is capturing complex temporal patterns, such as daily, weekly, and seasonal changes, while accounting for spatial differences across city zones. Traditional time series methods often have trouble with the non-linear relationships in urban mobility patterns, making deep learning methods useful for this problem.

This project focuses on developing robust deep learning models to forecast hourly taxi demand across 262 New York City zones. We implemented and compared four advanced architectures: Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), Temporal Convolutional Neural Networks (CNN), and Transformer models. Our approach emphasizes comprehensive feature engineering, automated hyperparameter optimization, and ensemble methods to maximize predictive accuracy.

The primary contribution of this work is a systematic comparison of deep learning architectures for taxi demand forecasting, demonstrating that ensemble methods combining multiple models significantly outperform individual architectures. We also provide insights into the

relative importance of different temporal features and the effectiveness of various optimization strategies.

II. DATASET DESCRIPTION

This study uses NYC Yellow Taxi trip data spanning October 2022 to January 2024, initially containing 38 million records and 19 features. The dataset was obtained from the NYC Taxi and Limousine Commission through NYC Open Data [1] and includes pickup/dropoff timestamps, locations, passenger counts, trip distances, and fare information.

A. Data Preprocessing

Our preprocessing had several essential steps. We removed 2.66 million bad records (7.0%) due to missing values, unrealistic trip times (over 3 hours), high fares (over \$200), and airport destinations that have different demand patterns. After filtering, we had 35.34 million valid trips from October 25, 2022, to January 3, 2024.

The key transformation involved aggregating individual trip records into hourly demand counts for each of the 262 pickup zones. This aggregation produced 781,196 hourly records with an average demand of 45.2 trips per hour per zone. The temporal coverage spans 374 unique days, providing sufficient data for robust model training and evaluation.

B. Exploratory Data Analysis

Our initial data analysis found clear time patterns important for forecasting. Daily demand shows strong daily cycles, dropping from 13,000 trips at midnight to the lowest levels around 5 AM, then rising to peak levels at 6 PM. Weekly patterns show Thursday has the highest demand, while Saturday has the lowest. Most trips (over 80%) are short-distance (under 2.5 miles) and single-passenger, reflecting typical urban commuting patterns.

The 262 zones show very different demand levels, with Manhattan zones having much higher demand than those in the outer regions. This difference across locations means we need zone-specific modelling and shows why location-based features are essential in our models.

III. METHODOLOGY

A. Feature Engineering

We built a feature engineering pipeline that creates 43 features from the raw data. Time features include hour-of-day, day-of-week, month, and holiday indicators, encoded using binary and cyclical encodings to capture repeating patterns. Lag features use historical demand at 1, 2, 3, 6, 12, 24, 48, and 168 hours (one week) to capture short- and long-term relationships.

Rolling statistics computed over 3, 6, 12, 24 and 48-hour windows provide moving averages and standard deviations that smooth noise and capture trend information. Location features include zone identifiers and spatial encodings to distinguish between different geographic areas with varying demand characteristics.

B. Baseline Models

To establish performance benchmarks, several baseline models were implemented. The Historical Average baseline uses the mean demand for each zone-hour combination from training data, achieving a test MAE of 11.01. Moving Average baselines with different window sizes (24, 48, 72, and 168 hours) were tested, with the 72-hour window performing best at MAE 62.56. Traditional regression models, including Linear Regression, Ridge Regression, and RandomForest, were implemented using the 43 engineered features, with RandomForest achieving the best performance with a test MAE of 5.56.

C. Model Architectures

We implemented four deep learning architectures optimized for time series forecasting [2]. The LSTM network uses two layers with 128 and 64 units and includes dropout regularization to prevent overfitting. The model takes 24-hour input sequences to predict the next hour's demand, using LSTM's ability to remember long-term patterns.

The GRU network uses the same two-layer structure as LSTM with 128 and 64 units for direct comparison. It uses fewer parameters per unit than LSTM while still capturing temporal patterns well. The simpler internal design makes GRU training faster while maintaining comparable performance.

The Temporal CNN model uses 1D convolutions with varying filter sizes to detect local temporal patterns. It has three convolutional layers with 64, 128, and 64 filters, followed by max-pooling and dropout layers.

The Transformer model uses multi-head attention with eight heads and 128-dimensional embeddings. This design captures long-range dependencies via attention mechanisms, enabling the model to focus on key time periods for predictions.

D. Hyperparameter Optimization

We used Optuna for automatic hyperparameter tuning, running 7-10 trials per model type [3]. The optimization used Tree-structured Parzen Estimator (TPE) sampling to efficiently search the hyperparameter space. For LSTM, optimal parameters included 96 and 160 units for the two layers, a dropout rate of 0.22, a learning rate of 0.0013, and a batch size of 512. For GRU, the optimal configuration used 64 and 96 units with similar dropout and batch size settings but a learning rate of 0.0006.

The optimization provided mixed results across architectures. LSTM performed worse after optimization, with MAE increasing from 6.98 to 7.08. GRU showed significant improvement, with MAE decreasing from 8.30 to 7.03. The Transformer also performed worse after optimization, with MAE increasing from 7.30 to 7.65.

E. Ensemble Methods

We implemented two ensemble approaches to combine individual model predictions. The weighted ensemble uses inverse validation MAE as weights. The weights were: LSTM (27.0%), GRU (26.9%), Transformer (24.5%), and RandomForest (21.6%). Note that the baseline RandomForest was not actually used in the ensemble methods, despite the weight calculation, but rather an alternative with a different feature space as input.

The stacked ensemble employs a LinearRegression meta-learner trained on validation predictions from the RandomForest and deep learning models. This approach achieved superior performance by learning optimal combination strategies, with meta-learner coefficients: GRU (0.397), LSTM (0.358), Transformer (0.180), and RandomForest (0.068) [4].

IV. RESULTS & DISCUSSION

A. Model Performance Comparison

Table I presents comprehensive performance metrics for the best-performing implemented models. RandomForest achieved the best overall performance with MAE of 5.56 trips/hour, RMSE of 13.66 trips/hour, and MAPE of 10.0%. Among deep learning models, the stacked ensemble performed best with an MAE of 6.96 trips/hour, representing a 25.2% performance gap compared to the RandomForest benchmark.

TABLE I. MODEL PERFORMANCE COMPARISON

Model	MAE	RMSE	MAPE	Gap To Best
RandomForest Baseline	5.56	13.66	10%	Best Model
Stacked Ensemble	6.96	14.85	38.5%	+25.2%
Weighted Ensemble	7.00	15.11	31.2%	+25.9%

Model	MAE	RMSE	MAPE	Gap To Best
GRU Optimized	7.03	15.19	39.9%	+26.4%
LSTM Optimized	7.08	15.22	28.1%	+27.3%
Transformer Optimized	7.65	16.79	31.5%	+37.6%

Individual deep learning models showed varying performance characteristics. LSTM and GRU achieved similar performance levels (MAE $\sim$ 7.0-7.1), confirming their effectiveness for sequential modelling. All deep learning models significantly outperformed simple baselines such as the Historical Average and Moving Average methods. However, RandomForest significantly outperformed all deep learning approaches, achieving an MAE of 5.56, demonstrating that well-engineered features with tree-based methods can outperform complex neural networks. The Transformer model performed worst among deep learning models (MAE 7.65), possibly because the relatively short sequence length (24 hours) did not fully utilize its attention mechanisms.

B. Ensemble Method Analysis

The ensemble methods showed clear benefits over individual models. The stacked ensemble performed better than the weighted ensemble by learning optimal combination strategies. The meta-learner analysis reveals that GRU and LSTM receive the highest coefficients (0.397 and 0.358, respectively), while RandomForest and Transformer contribute lower coefficients (0.069 and 0.180, respectively). This indicates that recurrent architectures work best for this temporal modelling task.

Residual analysis from the results shows systematic bias reduction across ensemble methods. Individual models exhibited higher bias: GRU (+1.01 trips/hour) and LSTM (+0.88 trips/hour), while ensemble approaches showed lower bias: Weighted Ensemble (+0.69 trips/hour) and Stacked Ensemble (+0.26 trips/hour). The stacked ensemble's significantly lower systematic bias suggests better model calibration [5].

C. Feature Importance & Model Insights

RandomForest feature importance analysis reveals that engineered features significantly contribute to model performance. The top feature (feature_103) accounts for 74.7% of the total importance, while the following nine features account for the remaining 25.3%. This concentration suggests that specific temporal or spatial patterns dominate taxi demand predictions.

Error analysis shows that 88.1% of prediction variance comes from random noise, while trend components account for only 11.9%. This means taxi demand has a lot of random variation that even advanced models cannot

predict, which sets realistic expectations for forecasting accuracy.

D. Temporal Pattern Analysis

Hourly error analysis shows patterns in model performance. Prediction errors are usually higher during peak demand periods (morning and evening rush hours), with greater variation during these times. The models work best during stable demand periods, especially late at night and early in the morning, when demand patterns are more predictable.

Weekly patterns show consistent performance across days, with slightly higher errors on weekends when demand deviates from typical weekday commuting behaviour. These insights suggest potential for time-adaptive models that adjust predictions based on temporal context.

V. CONCLUSION & REMARKS

This study demonstrates that while deep learning methods perform reasonably well for urban taxi demand forecasting, traditional machine learning with comprehensive feature engineering can be superior. RandomForest achieved the best performance (5.56 MAE), outperforming all deep learning approaches by significant margins of 25-38%. Among deep learning methods, ensemble techniques provided the best results.

Key findings include: (1) RandomForest with engineered features outperformed all deep learning approaches by significant margins; (2) Among deep learning methods, ensemble approaches work better than individual models; (3) LSTM and GRU architectures perform similarly for this temporal modeling task; (4) Limited systematic hyperparameter optimization provides only modest improvements; (5) Feature engineering is crucial, as demonstrated by RandomForest's superior performance.

A. Future Work

Future research directions include: (1) Incorporation of external data sources such as weather conditions, special events, and real-time traffic information; (2) Spatial modelling approaches that capture geographic dependencies between zones; (3) Hybrid approaches combining tree-based methods with neural networks; (4) Online learning methods for real-time model adaptation to changing demand patterns.

B. Limitations

This study has several limitations. The dataset covers only 15 months, potentially missing longer seasonal patterns. The study focuses exclusively on Yellow Taxi data, excluding other transportation modes that might improve prediction accuracy. The 24-hour input sequence

length may not be optimal for capturing weekly patterns, suggesting the need for further investigation with longer temporal sequences. Additionally, due to limited computational resources, hyperparameter optimization was limited to 7-10 trials per model, potentially failing to explore the optimal parameter space for each architecture fully.

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Appendix

Group Member Contributions

Mohammed Alamen Qassab (251285296): Project leadership and coordination, dataset discovery and acquisition, model tuning system implementation, ensemble method development, presentation creation and design, results analysis, and report writing.

Evan Kale Romano (251327329): Deep learning model architecture design and implementation, including LSTM, GRU, Temporal CNN, and Transformer models, baseline model development, including RandomForest and regression methods, model training pipeline creation and report writing.

Joseph Dominique Toma (251283541): Data preprocessing and cleaning pipeline development, dataset filtering and validation, feature engineering implementation, hourly aggregation and temporal split creation, data quality assessment, presentation content contributions, and report writing.

Ahmed El Shareif (251199128): Model evaluation framework development, error analysis and residual analysis implementation, performance metrics calculation, model comparison studies, validation protocol design, literature review, presentation enhancement, and report writing.